

Architectural Particular Specification

APPROVED SHOP DRAWINGS, AS-BUILT DRAWINGS, OPERATION AND MAINTENANCE MANUALS

SECTION 1.2

Rev	Date	Description
0	14 APR 2022	First Issue

SECTION 1.2

APPROVED SHOP DRAWINGS, AS-BUILT DRAWINGS, OPERATION AND MAINTENANCE MANUALS AND RECORDS

PART 1 GENERAL

1.01 SUMMARY

A. Section Includes:

1. Requirements of approved Shop Drawings, As-built Drawings and Documents
2. Pro forma structure and contents of Operation and Maintenance Manuals

B. Related Sections:

1. Section 1.5 Submittals Procedures
2. Special Conditions of Contract, and Specification - Preliminaries.

1.02 REQUIREMENTS OF APPROVED SHOP DRAWINGS, AS-BUILT DRAWINGS AND DOCUMENTS FOR ARCHITECTURAL WORKS

The Contractor shall refer the Particular Specification for the submission of as-built drawings / documents / any other information / operation and maintenance manual before the Substantial Completion, including but not limited to the following and any other completed works as instructed by the Architect. The Contractor is deemed to have allow time and cost to cope with any as-built records that may require.

Items of Work	Document / Information
Master layout, building and hard landscaping works	Final version of as-built drawings and maintenance manual
Proprietary roofing system	Final version of approved shop drawings and maintenance manual
Door and ironmongery	Final version of approved shop drawings and maintenance manual
Window, louvre including the de-mountable louvre, metal grille, and all types of shutters	Final version of approved shop drawing and O&M manual
Soft landscaping	Final version of as-built record and maintenance manual
Signage, proprietary expansion joint system	Final version of approved shop drawings
Turntable and double deck parking system	Final version of approved shop drawings and O&M manual
Turnstile, road blocker, portable x-ray and metal detector	Final version of approved shop drawings and O&M manual

1.03 PRO FORMA STRUCTURE AND CONTENTS OF OPERATION AND MAINTENANCE MANUALS

Contents of Operation and Maintenance Manual: -

Provide separate volume of Operation and Maintenance (O&M) Manuals for each installations/ services/ equipment: -

A. PROJECT INFORMATION

The project information shall include: -

1. Project title, and site address
2. Contract no., contract title, contractor/sub-contractor name, contact persons and telephones/faxes
3. Key contract dates: contract commencement, substantial completion, length of maintenance period.

B. SYSTEM DESCRIPTION

Provide detailed description of the system/services/equipment installed. It should include: -

1. Type(s) of system(s) and equipment installed
2. Locations of the system and major equipment, and what and where they serve
3. Description of operation and functions of the system and equipment
4. Design criteria, design data and parameters
5. General operating conditions, expected performance and energy and resources consumption where applicable

C. EQUIPMENT SCHEDULE

Schedule of all items of equipment and plant stating the location, name, model no., manufacturer's serial or reference numbers, duties and performance data.

D. MANUFACTURERS' CERTIFICATES / GUARANTEES / WARRANTIES AND STATUTORY INSPECTION CERTIFICATE

This shall include manufacturers' certificates such as factory test certificate, laboratory test reports, guarantees and warranties etc.

Any others statutory inspection certificate where required. Whilst photocopies may be allowed in the draft version, original should be included in the final version of the manuals.

E. SAFETY PRECAUTIONS FOR OPERATION AND MAINTENANCE

Describe in this section the hazards and safety precautions of which the operation and maintenance staff need to be aware. These should include :-

1. any known feature or operational characteristics of the equipment or systems installed which may produce a hazard,
2. any known hazards against which protection can be provided,
3. any mandatory requirements relating to safety,
4. any other safety precautions which should be observed,
5. any other relevant warnings.

F. OPERATION INSTRUCTIONS

Provide instructions for the safe and efficient operation, under both normal and emergency conditions, of the system installed. These will be in addition to the manufacturers' literature for equipment and plant items and shall include :-

1. an outline of the general operating mode;
2. control data (location, effect, object, sequence, limits of capability, modes, set points);
3. procedures and sequences for start-up, running and shut-down, under both normal and emergency conditions;
4. interlocks between equipment items;
5. operating procedures for stand-by equipment;
6. precautions necessary to overcome known hazards;
7. the means by which any potentially hazardous equipment may be made safe;
8. target figures for both energy consumption and energy costs;
9. forms for recording plant running hours, energy consumption and energy costs.

G. MAINTENANCE

1. Maintenance Instructions

Provide the manufacturers and the contractor/subcontractor's recommendations and instructions for maintenance for each item of plant and equipment installed. Clear distinction should be made between planned tasks (preventive maintenance) and work done on a corrective basis (corrective maintenance). Instructions shall be given on each of the following, as appropriate:-

- i. isolation and return to service of plant and equipment;
- ii. adjustments, calibration and testing;

- iii. dismantling and re-assembly;
- iv. exchange of components and assemblies;
- v. dealing with hazards which may arise during maintenance;
- vi. nature of deterioration, and the defects to be looked for;
- vii. special tools, test equipment and ancillary services.

H. RECORD DRAWINGS

1. Within 6 weeks after completion of works, the Contractor shall submit full sets of as-built records in the following copies:-
 - i. Hard Copies (Three Sets, A1 or A2 size subject to review with the Employer)
 - ii. Soft Copy (Three Sets*)
 - iii. Scanned copy of all hard-copies (Three Sets*)
 - iv. Exact number subject to verification and agreement with the Employer
2. The Submission of all of the above must be made no later than at the time of the handover of Works.
3. Provide a complete list of as-built drawings with drawing number/reference and brief description
4. Electronic files in latest version of AutoCAD format of all as-built drawings shall be submitted and archived in CD-ROM disks.
5. Scanned electronic files in TIFF or PDF or equivalent format of all as-built drawings with electronic drawing index files (shown with drawing number, drawing titles/description, scanned file names, etc.) in EXCEL format with hyperlink cross-reference function to the scanned electronic files shall be submitted and archived in CD-ROM disks.
6. As-Built drawings shall include relevant Working Drawings, Final approved Shop Drawings including approved test reports of any test required by the specifications and manufacturer shop drawings, Combined and Co-ordinated Services Drawings, Combined and Co-ordinated Services Builders Works Drawings and details which shall be issued to the satisfaction of the Architect/Engineer. The exact scope and details to be shown on each of the drawings shall be agreed with the Architect prior to the Date of Practical Completion.
7. Main Contractor shall also be responsible for submission of all as-built drawings, record drawings, test reports and certificates required under nominated sub contracts.

I. WARRANTY

This part shall be read in conjunction with the Special Conditions of Contract, Specification – Preliminaries and Appendix, and Particular Specification for the list of material/ system requiring joint written guarantees.

The Contractor, jointly with the Applicator and/or Manufacturer where applicable and as appropriate, shall submit a written guarantee for the materials, workmanship and performance of the proposed system(s) to the Architect for retention. No bulk materials and systems ordering work shall proceed prior to the Architect's confirmation of no-adverse-comment or review-without-objection of the warranty. It is the contractor's sole responsibility to ensure that the required materials or systems can be ordered according to the contractor's master programme. No claim for any additional time or cost due to any delayed submission of a satisfactory guarantee will be entertained.

J. TECHNICAL LITERATURE

A complete set of all manufacturers' literature should be provided for the plant and equipment installed, and assembled for the system. The literature shall apply only to specific items actually supplied for the contract.

The literature should provide the following information where applicable :-

1. description of the product as purchased with highlighted model numbers.
2. performance - behavioural characteristics of the equipment in use;
3. applications - suitability for use;
4. factory test reports, detailed drawings, circuit diagrams ;
5. operation and maintenance instructions details:
6. resources of plant, material and space required;
7. methods of operation and control;
8. cleaning and maintenance requirements;
9. spare parts lists ;
10. protective measures and safety precautions;
11. public safety considerations.

PART 2 PRODUCTS

(NOT USED)

PART 3 EXECUTION

(NOT USED)

END OF SECTION